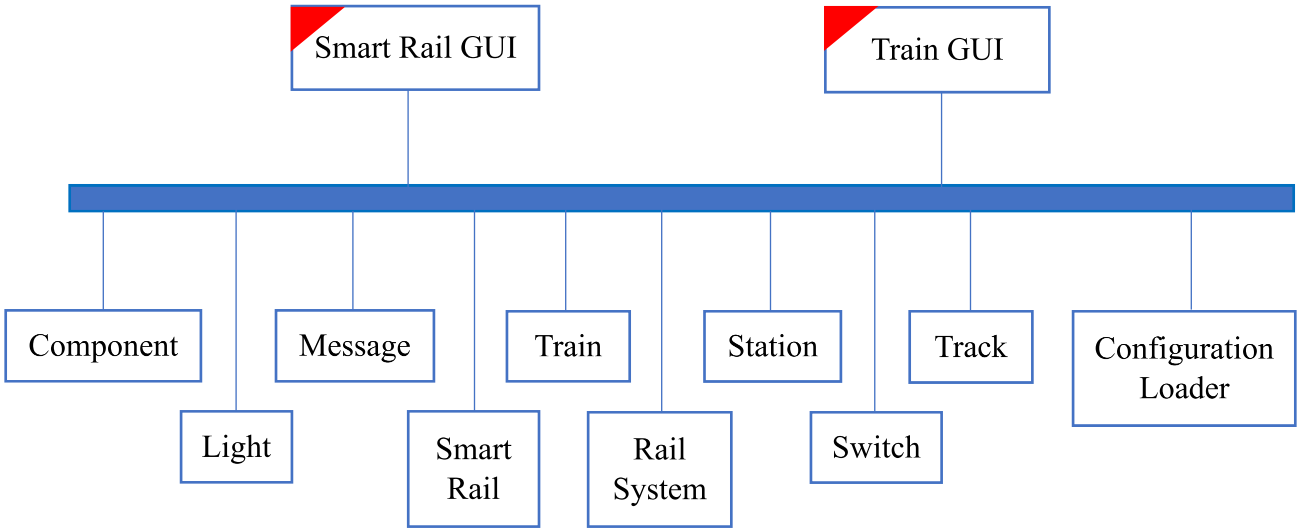
Proposed Design



Trigger: update from mouse

Description of the Design

Smart Rail GUI:

- Data:
  - Reference to Rail System
  - User input (source/destination stations, add train)
  - Animation/rendering details (train locations, lock states, etc.)
- Behavior:
  - Initialize and display the GUI
  - Handle user interactions (station selection, train addition)
  - Render the rail network and train movements
  - Communicate with RailSystem to initiate and update the simulation

Rail System:

- Data:
  - Collections of stations, switches, tracks, and trains
  - Overall simulation state
- Behavior:
  - Initialize the rail network based on configuration
  - Manage the simulation logic, including train movements and path finding
  - Coordinate interactions between trains and rail components
  - Handle events like train arrival, lock requests, and path conflicts

Configuration Loader:

- Data:
    - Configuration file contents
    - Parsed rail network components
- Behavior:
    - Load the rail network configuration from a file
    - Validate the configuration for correctness (no conflicts, dangling components, etc.)
    - Create the initial rail network components

Component (abstract):

- Data:
    - Position (x, y coordinates)
    - Locked state
    - Occupying train (if any)
- Behavior:
    - Process messages from neighboring components
    - Handle locking/unlocking requests
    - Notifying neighbors of changes in state

Station:

- Data:
    - Name
    - Locked state
    - Occupying train (if any)
- Behavior:
    - Process path finding and lock requests from trains
    - Handle train arrivals and departures

Switch:

- Data:
    - Locked state
    - Main and alternative track connections
    - Current switch position (main or alternative)
- Behavior:
    - Process path finding and lock requests from trains
    - Update switch position based on train movement

Track:

- Data:
    - Start and end coordinates
    - Number of segments
    - Locked state
    - Occupying train (if any)
- Behavior:
    - Process path finding and lock requests from trains
    - Handle train movement along the track

Train:

- Data:
    - Current position (x, y coordinates)
    - Current station
    - Destination station
    - Current path
    - Current path index
    - Lock retry count
    - Status (idle, seeking path, locking path, moving, exited)
- Behavior:
    - Find a path to the destination
    - Request locks along the path
    - Move along the secure path
    - Handle lock failures and retry path finding
    - Update GUI with current status